HCIA-AI V4.0 Certification Practice Exam • Final Mock Exam

Module 7: Comprehensive Final Mock Exam (100 Questions & Answer Key)

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Question 1

TRUE OR FALSE • 16 Points

The emergence of foundation models represents a shift from developing isolated single-task models to pre-training a general model that adapts across diverse downstream tasks.

A. True
B. False

Question 2

TRUE OR FALSE • 16 Points

Speech processing is a computer vision technology that only processes visual photographs of lips without analyzing audio soundwaves.

A. True
B. False

Question 3

TRUE OR FALSE • 16 Points

DeepSeek models use Multi-Head Latent Attention (MLA) architecture primarily to compress KV cache memory overhead during high-concurrency inference.

A. True
B. False

Question 4

TRUE OR FALSE • 16 Points

In supervised machine learning, the training dataset contains input feature vectors without any corresponding target label values.

A. True
B. False

Question 5

TRUE OR FALSE • 16 Points

L1 regularization drives certain weight coefficients strictly to zero, effectively performing embedded feature selection.

A. True
B. False

Page Answers: **Q1**: True • **Q2**: False • **Q3**: True • **Q4**: False • **Q5**: True

Question 6

TRUE OR FALSE • 16 Points

In hyperparameter optimization, exhaustive grid search is always much faster and computationally cheaper than random search when exploring high-dimensional spaces.

A. True
B. False

Question 7

TRUE OR FALSE • 16 Points

In deep networks using the Sigmoid activation function, gradients may diminish exponentially towards zero within five layers, causing severe vanishing gradients.

A. True
B. False

Question 8

TRUE OR FALSE • 16 Points

The Tanh activation function outputs values in the range (-1, 1) and completely eliminates the vanishing gradient problem in arbitrarily deep networks.

A. True
B. False

Question 9

TRUE OR FALSE • 16 Points

Batch Normalization standardizes intermediate layer inputs across mini-batches, stabilizing internal covariate shift and accelerating convergence.

A. True
B. False

Question 10

TRUE OR FALSE • 16 Points

During testing/evaluation mode, Dropout continues to randomly set 50% of neuron outputs to zero.

A. True
B. False

Page Answers: **Q6**: False • **Q7**: True • **Q8**: False • **Q9**: True • **Q10**: False

Question 11

TRUE OR FALSE • 16 Points

In Convolutional Neural Networks, the pooling layer introduces learnable weight parameters that are updated via backpropagation.

A. True
B. False

Question 12

TRUE OR FALSE • 16 Points

In LSTM networks, the Forget Gate determines the learning rate of the gradient descent optimizer.

A. True
B. False

Question 13

TRUE OR FALSE • 16 Points

PyTorch adopts a dynamic computational graph (Define-by-Run) where the graph is constructed on the fly as operations execute.

A. True
B. False

Question 14

TRUE OR FALSE • 16 Points

In PyTorch, calling optimizer.zero_grad() is necessary before loss.backward() because gradients accumulate by default.

A. True
B. False

Question 15

TRUE OR FALSE • 16 Points

In PyTorch, nn.CrossEntropyLoss requires the model output to be passed through an explicit torch.sigmoid() function first when performing multi-class classification.

A. True
B. False

Page Answers: **Q11**: False • **Q12**: False • **Q13**: True • **Q14**: True • **Q15**: False

Question 16

TRUE OR FALSE • 16 Points

In Data Parallelism, the entire training dataset is kept on one single device while neural network weights are replicated across all devices without batch partitioning.

A. True
B. False

Question 17

TRUE OR FALSE • 16 Points

In Pipeline Parallelism, the weight parameters within each individual layer are partitioned across multiple GPUs.

A. True
B. False

Question 18

TRUE OR FALSE • 16 Points

The core idea of Adapter Tuning is to insert small trainable adapter sub-modules into fixed layers of a pre-trained model while freezing backbone weights.

A. True
B. False

Question 19

TRUE OR FALSE • 16 Points

Huawei's Qiankun Intelligent Driving solution provides full-stack autonomous driving capabilities covering urban roads, highways, and automated valet parking.

A. True
B. False

Question 20

TRUE OR FALSE • 16 Points

In AI4Science, Huawei's Pangu-Weather model achieves weather forecasting speeds thousands of times faster than traditional numerical weather prediction (NWP).

A. True
B. False

Page Answers: **Q16**: False • **Q17**: False • **Q18**: True • **Q19**: True • **Q20**: True

Question 21

TRUE OR FALSE • 16 Points

In the hierarchical relationship among AI disciplines, Deep Learning is a subset of Machine Learning, and Machine Learning is a subset of Artificial Intelligence.

A. True
B. False

Question 22

TRUE OR FALSE • 16 Points

Connectionism is an AI school of thought inspired by biological neural networks and synaptic connection weights.

A. True
B. False

Question 23

TRUE OR FALSE • 16 Points

The three essential driving elements for modern Artificial Intelligence development are data, algorithms, and computing power.

A. True
B. False

Question 24

TRUE OR FALSE • 16 Points

Image classification tasks output both the bounding box spatial coordinates and the category label for each detected item.

A. True
B. False

Question 25

TRUE OR FALSE • 16 Points

Without considering regularization terms, the support vectors of a Support Vector Machine (SVM) are composed of the points closest to the separating hyperplane.

A. True
B. False

Page Answers: **Q21**: True • **Q22**: True • **Q23**: True • **Q24**: False • **Q25**: True

Question 26

SINGLE-ANSWER QUESTION • 16 Points

A computer uses labeled images to learn and distinguish between cats and dogs. What type of machine learning does this scenario represent?

- **A.** Clustering
- **B.** Reinforcement learning
- **C.** Unsupervised learning
- **D.** Supervised learning

Question 27

SINGLE-ANSWER QUESTION • 16 Points

Which of the following statements is TRUE regarding classification and regression in supervised learning?

- **A.** Both classification and regression predict discrete categorical labels
- **B.** Classification predicts discrete categorical labels, whereas regression predicts continuous numerical values
- **C.** Classification predicts continuous numerical values, whereas regression predicts discrete categories
- **D.** Both classification and regression predict continuous numerical values

Question 28

SINGLE-ANSWER QUESTION • 16 Points

According to the HCIA-AI V4.0 Model File specification, which model format uses Protocol Buffers to save parameter values by default so interrupted training can be resumed or fine-tuned?

- **A.** Uncompressed Bitmap (.bmp)
- **B.** Checkpoint (.ckpt)
- **C.** Comma-Separated Values (.csv)
- **D.** Plain Text Log (.txt)

Question 29

SINGLE-ANSWER QUESTION • 16 Points

When an overly complex model achieves 99.8% training accuracy but only 55% testing accuracy, it is suffering from:

- **A.** Vanishing Gradient
- **B.** High Bias
- **C.** Underfitting
- **D.** Overfitting

Question 30

SINGLE-ANSWER QUESTION • 16 Points

What is the harmonic mean of Precision and Recall commonly known as?

- **A.** R-Squared
- **B.** Accuracy
- **C.** F1-Score
- **D.** ROC Curve

Page Answers: **Q26**: D • **Q27**: B • **Q28**: B • **Q29**: D • **Q30**: C

Question 31

SINGLE-ANSWER QUESTION • 16 Points

Which distance metric measures the straight-line distance between two points in spatial geometry?

- **A.** Euclidean Distance
- **B.** Manhattan Distance
- **C.** Cosine Distance
- **D.** Hamming Distance

Question 32

SINGLE-ANSWER QUESTION • 16 Points

Which splitting criterion is used by the CART algorithm for classification trees?

- **A.** Information Gain Ratio
- **B.** Information Gain
- **C.** Gini Impurity
- **D.** Mean Squared Error

Question 33

SINGLE-ANSWER QUESTION • 16 Points

During neural network training, which of the following values are continuously updated by optimization algorithms using gradient descent?

- **A.** Model parameters
- **B.** Sample labels
- **C.** Learning rate
- **D.** Batch size

Question 34

SINGLE-ANSWER QUESTION • 16 Points

What is the value range of the standard ReLU activation function?

- **A.** [0, +inf)
- **B.** (-1, +1)
- **C.** (0, +1.0)
- **D.** (-inf, 0)

Question 35

SINGLE-ANSWER QUESTION • 16 Points

Which activation function is defined as f(x) = max(0, x) and helps reduce vanishing gradients in positive regions?

- **A.** Tanh
- **B.** Sigmoid
- **C.** ReLU
- **D.** Softsign

Page Answers: **Q31**: A • **Q32**: C • **Q33**: A • **Q34**: A • **Q35**: C

Question 36

SINGLE-ANSWER QUESTION • 16 Points

If a 32 x 32 feature map is processed through a 2 x 2 pooling layer with stride = 2 and no padding, what is the output size?

- **A.** 30 x 30
- **B.** 64 x 64
- **C.** 32 x 32
- **D.** 16 x 16

Question 37

SINGLE-ANSWER QUESTION • 16 Points

If a 32 x 32 image is convolved with a 5 x 5 kernel using stride = 1 and padding = 0, what is the output spatial dimension?

- **A.** 30 x 30
- **B.** 28 x 28
- **C.** 27 x 27
- **D.** 32 x 32

Question 38

SINGLE-ANSWER QUESTION • 16 Points

Which optimizer uses both the first moment (momentum) and the second moment (RMSProp) with bias corrections?

- **A.** Adam
- **B.** Newton-Raphson
- **C.** AdaGrad
- **D.** Standard SGD

Question 39

SINGLE-ANSWER QUESTION • 16 Points

What is the primary motivation behind the Residual Connection (skip connection) introduced in ResNet?

- **A.** Enabling identity mappings F(x) + x so gradients flow smoothly, preventing network degradation
- **B.** Replacing non-linear activation functions with direct linear identity pass-through layers
- **C.** Doubling total parameter counts across all hidden layers to maximize model capacity
- **D.** Eliminating the requirement for dedicated GPU accelerators during forward inference passes

Question 40

SINGLE-ANSWER QUESTION • 16 Points

Why is Positional Encoding essential in the Transformer architecture?

- **A.** To project sparse token index embeddings into low-dimensional normalized spatial coordinates
- **B.** To replace dense feedforward projection layers with fixed sinusoidal trigonometric operations
- **C.** To calculate cross-entropy loss gradients across multi-token target vocabulary dictionaries
- **D.** To provide the permutation-invariant self-attention mechanism with sequential order awareness

Page Answers: **Q36**: D • **Q37**: B • **Q38**: A • **Q39**: A • **Q40**: D

Question 41

SINGLE-ANSWER QUESTION • 16 Points

In the LSTM cell, which gate is responsible for deciding what percentage of information from the previous cell state should be discarded?

- **A.** Input Gate
- **B.** Reset Gate
- **C.** Forget Gate
- **D.** Output Gate

Question 42

SINGLE-ANSWER QUESTION • 16 Points

What does PEFT stand for in foundation model fine-tuning?

- **A.** Parameter-Efficient Fine-Tuning
- **B.** Pre-trained Exact Feature Transfer
- **C.** Parallel-Executed Forward Training
- **D.** Performance-Enhanced Formula Tool

Question 43

SINGLE-ANSWER QUESTION • 16 Points

Which PEFT method freezes base weights and injects trainable low-rank decomposition matrices into attention layers?

- **A.** Gradient descent
- **B.** Full retraining
- **C.** LoRA
- **D.** Cross validation

Question 44

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is NOT an AI deep learning framework?

- **A.** MindSpore
- **B.** MySQL
- **C.** PyTorch
- **D.** TensorFlow

Question 45

SINGLE-ANSWER QUESTION • 16 Points

In PyTorch, which module is specifically responsible for automatic differentiation?

- **A.** torch.distributed
- **B.** torch.cuda.stream
- **C.** torch.autograd
- **D.** torch.multiprocess

Page Answers: **Q41**: C • **Q42**: A • **Q43**: C • **Q44**: B • **Q45**: C

Question 46

SINGLE-ANSWER QUESTION • 16 Points

Which method in PyTorch calculates the backward gradients of a loss scalar with respect to leaf tensors?

- **A.** loss.backward()
- **B.** optimizer.step()
- **C.** tensor.zero_grad()
- **D.** model.forward()

Question 47

SINGLE-ANSWER QUESTION • 16 Points

Which method in PyTorch updates model parameters based on their computed gradients?

- **A.** optimizer.step()
- **B.** loss.backward()
- **C.** torch.cuda.empty_cache()
- **D.** model.eval()

Question 48

SINGLE-ANSWER QUESTION • 16 Points

What is the recommended PyTorch method for saving only the learned parameters (weights and biases) of a model?

- **A.** json.dump(model)
- **B.** pickle.dump(model)
- **C.** torch.save(model.state_dict(), PATH)
- **D.** torch.save(model, PATH)

Question 49

SINGLE-ANSWER QUESTION • 16 Points

Which model compression method converts 32-bit floating-point weights into 8-bit integers to reduce memory and accelerate inference?

- **A.** Knowledge Distillation
- **B.** Data Augmentation
- **C.** Model Quantization
- **D.** Model Pruning

Question 50

SINGLE-ANSWER QUESTION • 16 Points

What role does CANN play in Huawei's Ascend AI software stack?

- **A.** A mobile operating system kernel tailored specifically for consumer wearable smart devices
- **B.** The computing architecture and chip enablement layer interfacing with Ascend NPUs
- **C.** A relational database engine managing structured transactional business intelligence queries
- **D.** A cloud-based web dashboard designed for visualizing multi-node distributed training metrics

Page Answers: **Q46**: A • **Q47**: A • **Q48**: C • **Q49**: C • **Q50**: B

Question 51

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is Huawei's lightweight AI engine for ultra-fast, low-latency inference on mobile and IoT edge devices?

- **A.** Kunpeng OS
- **B.** MindSpore Lite
- **C.** GaussDB
- **D.** ModelArts Cloud

Question 52

SINGLE-ANSWER QUESTION • 16 Points

Within the broad landscape of foundation models, what is the primary purpose of Supervised Fine-Tuning (SFT)?

- **A.** Initializing model weights from scratch using uncurated random Gaussian noise distributions
- **B.** Adapting pre-trained models to follow specific user instructions and conversational formats
- **C.** Replacing deep neural attention layers with traditional rule-based scripts
- **D.** Pruning ninety percent of all weight parameters to minimize embedded deployment memory footprint

Question 53

SINGLE-ANSWER QUESTION • 16 Points

Which distributed training strategy partitions sequential layers of a deep neural network across multiple accelerators like stages in an assembly line?

- **A.** Zero-Shot Prompting
- **B.** Data Parallelism
- **C.** Pipeline Parallelism
- **D.** K-Means Clustering

Question 54

SINGLE-ANSWER QUESTION • 16 Points

What is the primary function of Retrieval-Augmented Generation (RAG)?

- **A.** Querying external knowledge stores to inject factual context into prompts, reducing hallucination
- **B.** Compiling high-performance native device driver modules for modern operating system kernels
- **C.** Deleting outdated customer database records automatically based on temporal retention rules
- **D.** Training foundational transformer base models from scratch using random initialization weights

Question 55

SINGLE-ANSWER QUESTION • 16 Points

In the RLHF workflow, which reinforcement learning algorithm is commonly used to optimize the language policy?

- **A.** Proximal Policy Optimization
- **B.** K-Means Centroid Clustering Algorithm
- **C.** Information Gain Ratio Decision Tree
- **D.** Ordinary Least Squares Linear Regression

Page Answers: **Q51**: B • **Q52**: B • **Q53**: C • **Q54**: A • **Q55**: A

Question 56

SINGLE-ANSWER QUESTION • 16 Points

What security threat occurs when an adversary crafts adversarial prompts to bypass safety guardrails and elicit dangerous content?

- **A.** Batch Normalization
- **B.** Weight Decay
- **C.** Cross-Validation
- **D.** Jailbreaking

Question 57

SINGLE-ANSWER QUESTION • 16 Points

Huawei's smart voice assistant across the HarmonyOS ecosystem is named:

- **A.** Siri
- **B.** Celia
- **C.** Alexa
- **D.** Bixby

Question 58

SINGLE-ANSWER QUESTION • 16 Points

Which sensor in autonomous driving emits laser pulses to construct high-precision 3D point cloud maps of the environment?

- **A.** Ultrasonic sensor
- **B.** Optical camera
- **C.** Millimeter-wave radar
- **D.** LiDAR

Question 59

SINGLE-ANSWER QUESTION • 16 Points

When an e-commerce platform launches a new product with zero user ratings or interactions, which problem does it encounter?

- **A.** Pipeline Stall
- **B.** Cold Start Problem
- **C.** Catastrophic Forgetting
- **D.** Vanishing Gradient

Question 60

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following core technologies allow mobile robots and autonomous vehicles to map unknown environments and localize themselves? (Multiple Choice)

- **A.** Simultaneous Localization and Mapping
- **B.** Visual Odometry using camera sensor fusion
- **C.** LiDAR point cloud scan matching algorithms
- **D.** Static geometric CAD blueprint querying without sensor feedback

Page Answers: **Q56**: D • **Q57**: B • **Q58**: D • **Q59**: B • **Q60**: ABC

Question 61

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following AI schools of thought originated from bionics and environmental cybernetics respectively?

A. Symbolism
B. Connectionism
C. Positivism
D. Behaviorism

Question 62

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common Computer Vision tasks?

A. Object Detection
B. Optical Character Recognition
C. Image Classification
D. Semantic Segmentation

Question 63

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are core compute innovations introduced in DeepSeek-V3?

A. Multi-Head Latent Attention for efficient KV caching
B. Static Global Average Pooling Layers
C. DeepSeekMoE architecture with shared experts
D. Recurrent Hebbian Plasticity Optimization

Question 64

MULTIPLE-ANSWER QUESTION • 18 Points

In the HCIA-AI V4.0 AI business evaluation workflow, which of the following statements regarding Data Pre-Evaluation, Model Post-Evaluation, and Regression Metrics are TRUE?

A. Pre-Evaluation inspects dataset Quality, Diversity, and Balance/Fairness before model training to prevent bias
B. Post-Evaluation assesses the trained model on the test set using classification metrics or regression metrics
C. When extreme outliers are present in regression data, MAE is more robust than MSE and RMSE because it does not square errors
D. Manual Feature Engineering is mandatory for Computer Vision and NLP models rather than Recommendation Systems

Question 65

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following hyperparameters may be defined and tuned during machine learning training?

A. Learning rate
B. Number of epochs
C. Model weights and biases
D. Batch size

Page Answers: **Q61**: BD • **Q62**: ABCD • **Q63**: AC • **Q64**: ABC • **Q65**: ABD

Question 66

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements accurately reflect Feedforward Neural Network (FNN) properties and gradient stability mechanisms in HCIA-AI V4.0?

A. An FNN transmits signals unidirectionally with no feedback and can be represented as a Directed Acyclic Graph (DAG)

B. In an FNN neuron, the bias parameter 'b' represents the activation threshold

C. The Sigmoid activation function has a maximum derivative of 0.25 (1/4), contributing to vanishing gradients in deep networks

D. L1 and L2 weight regularization cause exploding gradients and must always be avoided during deep network training

Question 67

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following tasks belong to the data preprocessing stage?

A. Data deduplication

B. Feature normalization and standardization

C. Outlier handling

D. Missing value imputation

Question 68

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are supervised machine learning algorithms?

A. Support Vector Machines

B. K-Means Clustering

C. Multiple Linear Regression

D. Principal Component Analysis

Question 69

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following algorithms belong to Ensemble Learning?

A. Random Forest

B. AdaBoost

C. Single Perceptron

D. Gradient Boosted Decision Tree

Question 70

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following activation functions are smooth, bounded S-shaped (sigmoidal) curves commonly used in neural networks?

A. Sigmoid function

B. Hyperbolic Tangent or Tanh function

C. Standard Rectified Linear Unit

D. Leaky ReLU function

Page Answers: **Q66**: ABC • **Q67**: ABCD • **Q68**: AC • **Q69**: ABD • **Q70**: AB

Question 71

MULTIPLE-ANSWER QUESTION • 18 Points

In the Transformer attention mechanism, which vectors are multiplied via dot-product to compute raw compatibility attention weights?

- **A.** Value vector
- **B.** Query vector
- **C.** Key vector
- **D.** Residual shortcut vector

Question 72

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following algorithms are commonly used first-order gradient-based optimizers in deep learning?

- **A.** RMSProp
- **B.** Momentum SGD
- **C.** Adam
- **D.** K-Means

Question 73

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are foundational components of a Convolutional Neural Network (CNN)?

- **A.** Convolutional layer with learnable filter kernels
- **B.** Pooling layer
- **C.** Learnable positional encodings and multi-head cross-attention projections
- **D.** Non-linear activation functions

Question 74

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are primary techniques for preventing overfitting in deep neural networks?

- **A.** Dropout
- **B.** Early stopping
- **C.** L1 and L2 weight regularization
- **D.** Data augmentation

Question 75

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are key architectural advantages of the Transformer over classical RNN models?

- **A.** Full parallelization during training across sequence tokens
- **B.** Ability to scale efficiently to billions of parameters
- **C.** Eliminating the need for any training data
- **D.** Direct global dependency modeling without distance decay

Page Answers: **Q71**: BC • **Q72**: ABC • **Q73**: ABD • **Q74**: ABCD • **Q75**: ABD

Question 76

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about the Softmax activation function are correct?

- **A.** It maps arbitrary real-valued logits into a valid probability distribution
- **B.** The sum of all output probabilities across all classes equals exactly 1
- **C.** Every individual output probability value is strictly bounded between 0 and 1
- **D.** It guarantees that all output probabilities are strictly negative numbers

Question 77

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are primary functions provided by an AI Development Framework?

- **A.** Model optimization, serialization, and deployment tools
- **B.** High-level front-end programming APIs
- **C.** Hardware abstraction and computation scheduling
- **D.** Automatic differentiation engine

Question 78

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are key characteristics of Eager Execution (Define-by-Run) mode?

- **A.** Immediate execution of operations with direct numerical feedback
- **B.** Line-by-line interactive debugging using standard Python tools like pdb
- **C.** Whole-graph ahead-of-time memory reuse planning and optimization
- **D.** Immutability of execution topologies during runtime

Question 79

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following techniques directly compress numerical weight precision or model footprint in deep neural networks?

- **A.** Quantization
- **B.** Pruning
- **C.** Knowledge distillation
- **D.** Data augmentation

Question 80

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about PyTorch DataLoader are correct?

- **A.** It supports parallel multi-process data loading via the num_workers parameter
- **B.** It automatically batches individual data samples into mini-batch tensors
- **C.** It automatically searches and compiles optimal neural network architectures without user code
- **D.** It supports stochastic shuffling of dataset samples at the beginning of each training epoch

Page Answers: **Q76**: ABC • **Q77**: ABCD • **Q78**: AB • **Q79**: ABC • **Q80**: ABD

Question 81

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are major distributed parallelism modes for training large foundation models?

- **A.** Model pruning
- **B.** Tensor parallelism
- **C.** Data parallelism
- **D.** Pipeline parallelism

Question 82

MULTIPLE-ANSWER QUESTION • 18 Points

In the standard RLHF training workflow, which training stages follow Supervised Fine-Tuning (SFT)?

- **A.** Raw Web Data Scraping and Pre-tokenization
- **B.** Reward Model Training on human preference comparisons
- **C.** Policy Optimization using PPO
- **D.** Static C++ Kernel Compilation

Question 83

MULTIPLE-ANSWER QUESTION • 18 Points

A standard enterprise RAG system mainly consists of which core functional modules?

- **A.** Retrieval module
- **B.** Pre-training module
- **C.** Generation module
- **D.** Fine-tuning module

Question 84

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are common Prompt Engineering techniques?

- **A.** Chain-of-Thought Prompting
- **B.** Zero-Shot Prompting
- **C.** Role-Based Prompting
- **D.** Few-Shot In-Context Learning

Question 85

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are main safety risks associated with large model deployments?

- **A.** Guaranteed convergence to zero generalization error
- **B.** Sensitive data leakage from training memorization
- **C.** Hallucination of harmful or false advice
- **D.** Prompt Injection and Jailbreaking attacks

Page Answers: **Q81**: BCD • **Q82**: BC • **Q83**: AC • **Q84**: ABCD • **Q85**: BCD

Question 86

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following core technologies are integrated into an end-to-end Voice Assistant (e.g., Celia)?

- **A.** Dialogue Management
- **B.** Text-to-Speech
- **C.** Natural Language Understanding
- **D.** Photometric stereo surface normal reconstruction

Question 87

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are primary sensors used in autonomous driving perception systems?

- **A.** Optical HD Cameras
- **B.** Ultrasonic Sensors
- **C.** Millimeter-Wave Radar
- **D.** Light Detection and Ranging

Question 88

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are typical applications of AI in Smart Homes?

- **A.** Intelligent voice-activated appliance control
- **B.** Smart climate control and energy optimization
- **C.** Anomaly detection
- **D.** High-frequency automated stock market equity arbitrage

Question 89

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following represent the classic main types of recommendation algorithms?

- **A.** Collaborative Filtering
- **B.** Content-Based Filtering
- **C.** Random Permutation Shuffling
- **D.** Lexical Alphabetical Sorting

Question 90

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following components are part of Huawei's full-stack AI development ecosystem?

- **A.** ModelArts development platform
- **B.** Ascend AI processors
- **C.** CANN compute architecture
- **D.** MindSpore AI framework

Page Answers: **Q86**: ABC • **Q87**: ABCD • **Q88**: ABC • **Q89**: AB • **Q90**: ABCD

Question 91

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are recognized as the three foundational pillars powering the rapid advancement of modern AI?

- **A.** Massive high-quality training Data
- **B.** High-performance AI Computing Power
- **C.** Advanced Deep Learning and Machine Learning Algorithms
- **D.** Highly structured relational database tables with fixed SQL schemas

Question 92

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding Support Vector Machines (SVM) are correct?

- **A.** The objective of SVM is to find an optimal hyperplane that maximizes the margin
- **B.** Support vectors are the subset of training samples closest to the decision boundary
- **C.** Kernel functions allow SVM to map input vectors into higher-dimensional feature spaces
- **D.** In high-dimensional feature spaces, an optimal hyperplane is constructed to separate classes

Question 93

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, the method called on the optimizer to reset previously calculated gradients before computing new gradients is optimizer.____().

Answer: _____

Question 94

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, the method called on the scalar loss tensor to calculate gradients via backpropagation is loss.____().

Answer: _____

Question 95

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, the optimizer method that updates model parameters based on computed gradients is optimizer.____().

Answer: _____

Page Answers: **Q91**: ABC • **Q92**: ABCD • **Q93**: zero_grad • **Q94**: backward • **Q95**: step

Question 96

FILL-IN-THE-BLANK QUESTION • 16 Points

In Huawei MindSpore, the forward execution logic of a neural network module is implemented inside the def ____(self, x): method.

Answer: _____

Question 97

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, the method implemented inside an nn.Module subclass to define the forward computational graph is def ____(self, x):.

Answer: _____

Question 98

FILL-IN-THE-BLANK QUESTION • 16 Points

Implemented in C/C++, the standalone deployment tool ____ enables high-performance LLM inference locally or in the cloud without needing an AI framework.

Answer: _____

Question 99

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, the method that returns a dictionary containing all learnable model weights for saving to disk is model.____().

Answer: _____

Question 100

FILL-IN-THE-BLANK QUESTION • 16 Points

In PyTorch, to disable gradient tracking during model evaluation and save memory, code is executed inside the context with torch.____():.

Answer: _____

Page Answers: **Q96**: construct • **Q97**: forward • **Q98**: llama.cpp • **Q99**: state_dict • **Q100**: no_grad